Programme on **'Python Programming** ' Skillset

Programme Schedule Model:

- Total Mentor-led VILT topics-session Duration : 86 Hours
- Practice Duration: 180 hours
- Total Duration for self-paced Course(s): 12 hours
- Option-1 (**suggested**): 2.5 Hours per day; 5 days a week; Total: 8 weeks
- Option-2 (challenging with academic needs) : 4 Hours per day; 5 days a week; Total: 5 weeks

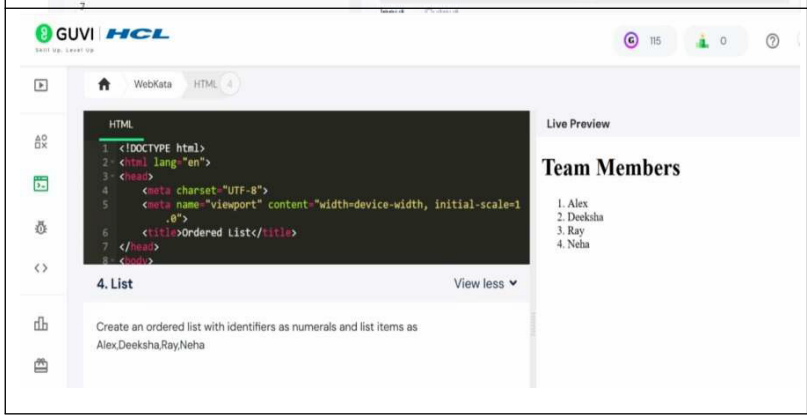
Instructor: Industry Mentors

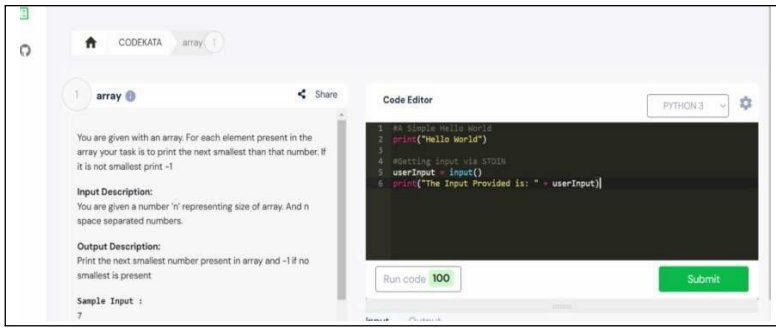
- Pre & Post Assessment (via AI-powered Assessment platform)
- Mock Assessments (1 each in the last 3 weeks of the programme)
- Session-wise Quiz challenges (for better interactions & knowledge sharing)
- Hands-on practice with use cases

Programme Deliverables:

- Consolidated Learners' performance report on each skillset
- Certificates for completed Participants
- ZEN Query portal for ad hoc doubt clarification by the participants, out of the session (TAT: same day; Clarification types: email/chat/gmeet)

GUVI's Exclusive Practice Platforms:

	Codekata : https://www.guvi.in/code-kata.html A tool-kit specifically developed to boost the coding skills and makes you ever-ready to crack interviews.
	**Webkata**: https://www.guvi.in/webkata.html A cloud-based module to hone your front end skills without any hassle of local environment setup.



IDE : https://www.guvi.in/ide/
GUVI IDE is an Integrated Development Environment that lets you write, edit, run, test & debug your code.

Table of Contents - Python Programming

1. **Introduction to Python** (Mentor-led Duration: 4 Hrs)
   - Overview of Python and Its Applications
   - Installing Python and Setting Up Environment (IDLE, VSCode)
   - Writing and Running Python Scripts
   - Understanding Python Syntax and Indentation
2. **Python Essentials** (Mentor-led Duration: 8 Hrs)
   - Variables and Data Types
   - Input/Output Operations
   - Operators (Arithmetic, Comparison, Logical, Assignment)
   - Comments and Code Readability
3. **Control Flow** (Mentor-led Duration: 14 Hrs)
   - Conditional Statements (if, elif, else)
   - Loops (for, while)
   - Loop Control Statements (break, continue, pass)
   - Match-case (structural pattern matching)
4. **Functions and Modules** (Mentor-led Duration: 3 Hrs)
   - Defining and Calling Functions
   - Function Arguments and Return Values
   - Importing and Using Modules (Built-in and Custom)
5. **Data Structures in Python** (Mentor-led Duration: 16 Hrs)
   - **Lists**
     - Creating and Manipulating Lists
     - List Comprehensions
   - **Working with Strings**
     - Creating Strings
     - String Concatenation
     - Multi-line Strings
     - String Formatting
     - Accessing Characters in a String
     - String slicing
     - String Methods
   - **Tuples**
     - Characteristics and Usage
   - **Dictionaries**
     - Key-Value Pair Manipulations
   - **Sets**

    - Unique Element Operations

6. **File Handling** (Mentor-led Duration: 2 Hrs)
   - Reading and Writing Files
   - Working with Different File Formats (Text, CSV)
   - Exception Handling in File Operations

7. **Object-Oriented Programming (OOP)** (Mentor-led Duration: 4 Hrs)
   - Introduction to OOP Concepts
   - Defining Classes and Objects
   - Constructors and Methods
   - Inheritance and Polymorphism

8. **Python Libraries and Tools** (Mentor-led Duration: 10 Hrs)
   - **Numpy**: Basics and Array Operations
   - **Pandas**: DataFrames and Series
   - **Matplotlib**: Basic Plotting
   - Introduction to Other Libraries (e.g., Requests, OS)

9. **Error Handling and Debugging** (Mentor-led Duration: 4 Hrs)
   - Types of Errors in Python
   - Try, Except, Finally Blocks
   - Debugging Techniques and Tools

10. **Unit Testing with Python** (Mentor-led Duration: 4 Hrs)
    - Organizing Your Tests With the TestCase Class
    - Creating Test Cases
    - Running unittest Tests
    - Exploring the Assert Methods

11. **Logging with Python** (Mentor-led Duration: 2 Hrs)
    - Python's Logging Module
    - Configuring Your Logging System
    - Common Logging Pitfalls and Solutions

12. **Advanced Topics** (Mentor-led Duration: 12 Hrs)
    - Lambda, Filters, Map & Reduce
    - Debugging in Python
    - Pickling
    - Python JSON
    - Python API and web scraping
    - Working with Python SQLite
    - Iterators and Generators
    - Decorators
    - Working with Regular Expressions
    - Introduction to Virtual Environments and Package Management (pip)

Harnessing Generative Artificial Intelligence (GenAI)

GitHub CoPilot – Utilization (3 Hours)

- GitHub Copilot Overview.
- Basic Copilot Usage.
- Understanding Copilot Suggestions.
- Generative Commit Messages.
- Understanding Copilot Chat.
- Copilot for the CLI.
- Copilot's Limitations and Strengths.
- Troubleshooting and Common Issues.
- GitHub Copilot Language Support
- Code Documentation.
- Copilot in IDEs.
- Code Optimization with Copilot.
- Refactoring Strategies.
- NLP and GitHub Copilot
- Debugging with Copilot.
- Copilot Best Practices.
- Usage of CoPilot in the existing/legacy code

Mini Project Development

- Business Use cases as Project Problem Statement.
- With Mentor-led doubts clarification support.